

Boss Challenge — Paper 05 (Class 7)

Transportation in Animals & Plants | Reproduction in Plants | Perimeter & Area

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. Blood is pumped to all parts of the body by the:
(A) heart
(B) kidney
(C) brain
(D) lungs
2. The reproductive part of a plant is the:
(A) root
(B) stem
(C) leaf
(D) flower
3. The area of a rectangle of length 8 cm and breadth 3 cm is:
(A) 24 cm²
(B) 11 cm²
(C) 24 cm
(D) 22 cm²
4. Blood vessels that carry blood AWAY from the heart are called:
(A) arteries
(B) capillaries
(C) veins
(D) nerves
5. In plants, the transfer of pollen from the anther to the stigma is called:
(A) pollination
(B) transpiration
(C) fertilisation
(D) germination
6. The perimeter of a rectangle of length 8 cm and breadth 3 cm is:
(A) 24 cm
(B) 11 cm
(C) 44 cm
(D) 22 cm
7. The red colour of blood is due to a substance in red blood cells called:
(A) platelets
(B) haemoglobin
(C) urea
(D) plasma

8. Which is an example of asexual reproduction in plants (no seeds/flowers needed)?

- (A) fertilisation in a flower
- (B) a mango seed sprouting
- (C) pollination by bees
- (D) growing a new rose plant from a cutting

9. The area of a square of side 5 cm is:

- (A) 20 cm^2
- (B) 25 cm^2
- (C) 25 cm
- (D) 10 cm^2

10. Blood vessels that carry blood back TOWARDS the heart are:

- (A) xylem
- (B) arteries
- (C) veins
- (D) capillaries

11. In a flower, the male part that produces pollen is the:

- (A) pistil
- (B) stamen (anther)
- (C) sepal
- (D) petal

12. The perimeter of a square of side 5 cm is:

- (A) 15 cm
- (B) 25 cm
- (C) 20 cm
- (D) 10 cm

13. In plants, water and minerals are carried UPWARD from the roots by the:

- (A) roots only
- (B) xylem
- (C) phloem
- (D) stomata

14. In a flower, the female part that receives pollen is the:

- (A) petal
- (B) pistil (stigma)
- (C) sepal
- (D) stamen

15. The area of a right triangle with base 6 cm and height 4 cm is:

- (A) 10 cm^2
- (B) 12 cm
- (C) 12 cm^2
- (D) 24 cm^2

16. The waste-filtering organs that clean the blood and make urine are the:

- (A) heart
- (B) liver
- (C) lungs
- (D) kidneys

17. The fusion of the male and female gametes in a flower is called:

- (A) pollination
- (B) fertilisation
- (C) transpiration
- (D) germination

18. 1 square metre (1 m^2) equals how many square centimetres?

- (A) 100 cm^2
- (B) $10,000 \text{ cm}^2$
- (C) $1,000 \text{ cm}^2$
- (D) $1,00,000 \text{ cm}^2$

19. Tiny blood cells that help blood to clot when we get a cut are the:

- (A) platelets
- (B) red blood cells
- (C) plasma
- (D) white blood cells

20. Growing a new plant from the leaf of a Bryophyllum (with tiny buds on its edges) is an example of:

- (A) vegetative (asexual) reproduction
- (B) pollination
- (C) seed reproduction
- (D) fertilisation

21. The area of a parallelogram with base 10 cm and height 4 cm is:

- (A) 20 cm^2
- (B) 14 cm^2
- (C) 28 cm^2
- (D) 40 cm^2

22. White blood cells (WBCs) in the body mainly:

- (A) fight germs and protect against disease
- (B) carry oxygen
- (C) clot blood
- (D) make urine

23. Spores are used for reproduction by:
- (A) mango trees
 - (B) roses from cuttings
 - (C) a hen
 - (D) ferns and fungi (like mushrooms)
24. A square has an area of 49 cm^2 . The length of its side is:
- (A) 7 cm
 - (B) 196 cm
 - (C) 49 cm
 - (D) 24.5 cm
25. The loss of water vapour from the leaves of a plant is called:
- (A) transpiration
 - (B) germination
 - (C) respiration
 - (D) photosynthesis
26. After fertilisation, the part of the flower that grows into a FRUIT is the:
- (A) petal
 - (B) ovary
 - (C) sepal
 - (D) stigma
27. The perimeter of a square whose area is 36 cm^2 is:
- (A) 12 cm
 - (B) 36 cm
 - (C) 24 cm
 - (D) 9 cm
28. Root hairs of a plant are useful because they:
- (A) carry food down
 - (B) release oxygen
 - (C) make food
 - (D) increase the surface area to absorb more water from soil
29. Pollination carried out with the help of insects like bees is called:
- (A) self fertilisation
 - (B) wind pollination
 - (C) germination
 - (D) insect pollination
30. The circumference of a circle of radius 7 cm is (use $\pi = 22/7$):
- (A) 22 cm
 - (B) 44 cm
 - (C) 49 cm
 - (D) 154 cm

31. Which carries digested FOOD to all parts of a plant?
- (A) stomata
 - (B) phloem
 - (C) xylem
 - (D) roots
32. Reproduction that needs only ONE parent and produces plants identical to it is:
- (A) sexual reproduction
 - (B) fertilisation
 - (C) pollination
 - (D) asexual reproduction
33. The area of a circle of radius 7 cm is (use $\pi = 22/7$):
- (A) 44 cm²
 - (B) 22 cm²
 - (C) 154 cm²
 - (D) 49 cm²
34. The rhythmic throbbing felt in an artery (e.g., at the wrist) with each heartbeat is the:
- (A) pulse
 - (B) breath
 - (C) nerve signal
 - (D) clot
35. Budding (a small bud growing out and detaching) is the way ____ reproduces.
- (A) a mango tree
 - (B) a human
 - (C) a fern
 - (D) yeast
36. A wire of length 24 cm is bent into a square. The side of the square is:
- (A) 8 cm
 - (B) 6 cm
 - (C) 12 cm
 - (D) 24 cm
37. Which part of the blood is the pale yellow liquid that carries cells and dissolved food?
- (A) haemoglobin
 - (B) urine
 - (C) platelets
 - (D) plasma
38. Seeds with wings or hair (like the drumstick or cotton) are usually dispersed by:
- (A) wind
 - (B) water
 - (C) explosion
 - (D) animals only

39. The area of a rectangle is 36 cm^2 and its length is 9 cm. Its breadth is:
- (A) 27 cm
 - (B) 324 cm
 - (C) 4 cm
 - (D) 45 cm
40. Excess water and salts are removed from our body as urine, and also through the skin as:
- (A) bile
 - (B) blood
 - (C) sweat
 - (D) oxygen
41. A potato grows new plants from the small 'eyes' on it. This is an example of:
- (A) vegetative propagation (from a tuber)
 - (B) spore formation
 - (C) seed reproduction
 - (D) pollination
42. Doubling the side of a square makes its area become:
- (A) 4 times as large
 - (B) 8 times
 - (C) the same
 - (D) 2 times
43. Capillaries are important because they:
- (A) pump blood
 - (B) are tiny vessels where blood exchanges oxygen and food with cells
 - (C) store urine
 - (D) make blood cells
44. Which flower part attracts insects for pollination?
- (A) the roots
 - (B) the green sepals
 - (C) the xylem
 - (D) colourful petals (and nectar/scent)
45. The area of a triangle is 20 cm^2 and its base is 5 cm. Its height is:
- (A) 16 cm
 - (B) 4 cm
 - (C) 8 cm
 - (D) 100 cm
46. Arteries have thick, elastic walls because they:
- (A) make blood
 - (B) carry blood slowly back
 - (C) store food
 - (D) carry blood pumped out at high pressure

47. Self-pollination means pollen lands on the stigma of:

- (A) the same flower (or same plant)
- (B) a different plant
- (C) an insect's body
- (D) a different species

48. The perimeter of a rectangle is 20 cm and its length is 6 cm. Its breadth is:

- (A) 4 cm
- (B) 2 cm
- (C) 14 cm
- (D) 8 cm

49. Which of these is part of the human excretory system?

- (A) heart
- (B) stomach
- (C) lung
- (D) kidney

50. Which is dispersed mainly by water?

- (A) a sticky burr
- (B) a pea from a bursting pod
- (C) a coconut
- (D) a dandelion seed

51. A square and a rectangle have the same perimeter of 24 cm. If the rectangle is 8 cm by 4 cm, the square's side is:

- (A) 4 cm
- (B) 8 cm
- (C) 6 cm
- (D) 12 cm

52. Blood carries oxygen from the lungs to the body because red cells contain haemoglobin, which:

- (A) makes urine
- (B) destroys germs
- (C) binds oxygen and releases it to the cells
- (D) clots blood

53. A flower that has both stamens and pistil is called a:

- (A) bisexual flower
- (B) root flower
- (C) seedless flower
- (D) unisexual flower

54. The area of the four walls aside, the floor of a square room of side 4 m has an area of:
- (A) 16 m²
 - (B) 12 m²
 - (C) 16 m
 - (D) 8 m²
55. Which animal has a simple transport system where blood is NOT always inside vessels (an 'open' system)?
- (A) a human
 - (B) a dog
 - (C) a fish
 - (D) an insect like a cockroach
56. Germination is the process in which a:
- (A) pollen reaches the stigma
 - (B) flower forms fruit
 - (C) leaf makes food
 - (D) seed grows into a new young plant
57. A rectangular field is 50 m long and 30 m wide. The cost of fencing it at ₹2 per metre is:
- (A) ₹160
 - (B) ₹3000
 - (C) ₹320
 - (D) ₹110
58. Veins have valves in them mainly to:
- (A) pump the blood
 - (B) stop blood from flowing backwards
 - (C) add oxygen
 - (D) clean the blood
59. Cross-pollination usually needs an agent such as:
- (A) wind, water or insects
 - (B) the kidneys
 - (C) sunlight only
 - (D) the roots
60. The area of a square of side 9 cm is:
- (A) 36 cm²
 - (B) 18 cm²
 - (C) 81 cm
 - (D) 81 cm²

61. Which is the correct path of blood?
- (A) heart → veins → body → arteries
 - (B) lungs → kidneys → heart
 - (C) heart → arteries → body → veins → heart
 - (D) stomach → heart → brain
62. Which structure protects the flower when it is still a bud?
- (A) anther
 - (B) sepals
 - (C) petals
 - (D) ovary
63. The breadth of a rectangle is half its length. If the length is 12 cm, its area is:
- (A) 36 cm²
 - (B) 72 cm²
 - (C) 144 cm²
 - (D) 18 cm²
64. The main job of the circulatory system is to:
- (A) think and learn
 - (B) carry oxygen, food and wastes around the body
 - (C) make food
 - (D) digest food
65. Sexual reproduction in plants results in seeds that:
- (A) are always identical to one parent
 - (B) may grow into plants with a mix of features from two parents
 - (C) never grow
 - (D) have no parents
66. A square lawn has a side of 10 m. A path is NOT involved; just its area is:
- (A) 1000 m²
 - (B) 100 m²
 - (C) 20 m²
 - (D) 40 m²
67. Plants do not have a heart. Water still rises to tall treetops mainly because of:
- (A) photosynthesis
 - (B) a pumping heart
 - (C) transpiration pull (water evaporating from leaves draws more up)
 - (D) roots pushing only
68. Which of these reproduces by FRAGMENTATION (breaking into pieces that each grow)?
- (A) yeast
 - (B) Spirogyra (a pond alga)
 - (C) a rose from a cutting
 - (D) a mango tree

69. The perimeter of a triangle with sides 5 cm, 6 cm and 7 cm is:
- (A) 105 cm
 - (B) 18 cm
 - (C) 9 cm
 - (D) 11 cm
70. An adult human heart beats about how many times per minute at rest?
- (A) about 7 times
 - (B) about 300 times
 - (C) about 72 times
 - (D) 0 times
71. The ovules inside the ovary become ____ after fertilisation.
- (A) petals
 - (B) roots
 - (C) seeds
 - (D) leaves
72. A rectangular garden 20 m by 15 m is surrounded by a 1 m wide path outside it. The outer length and breadth are:
- (A) 24 m and 19 m
 - (B) 20 m and 15 m
 - (C) 22 m and 17 m
 - (D) 21 m and 16 m
73. Which liquid waste do the kidneys produce by filtering the blood?
- (A) urine
 - (B) bile
 - (C) saliva
 - (D) sweat
74. Which is a difference between sexual and asexual reproduction in plants?
- (A) asexual always needs insects
 - (B) sexual gives identical copies
 - (C) both always use seeds
 - (D) sexual needs two parents/gametes; asexual needs only one parent
75. The area of a triangle with base 8 cm and height 5 cm is:
- (A) 40 cm²
 - (B) 20 cm²
 - (C) 13 cm²
 - (D) 20 cm

76. Blood is mainly made of cells plus a liquid. The liquid part is about ____ of blood.
- (A) a tiny amount
 - (B) none
 - (C) more than half (the major part)
 - (D) all of it
77. Why are seeds dispersed away from the parent plant?
- (A) so new plants get space, light and water without crowding
 - (B) to stop germination
 - (C) to make the plant taller
 - (D) to feed animals only
78. The side of a square whose perimeter is 28 cm is:
- (A) 14 cm
 - (B) 28 cm
 - (C) 7 cm
 - (D) 4 cm
79. Which best describes the role of blood vessels called capillaries?
- (A) they pump blood
 - (B) they connect arteries to veins and let exchange happen at cells
 - (C) they store urine
 - (D) they carry only air
80. A gardener wants many rose plants exactly like a prize rose. The best method is:
- (A) growing from seeds
 - (B) using spores
 - (C) waiting for wind pollination
 - (D) growing them from cuttings (vegetative propagation)
81. A floor 5 m by 4 m is to be covered with square tiles of side 1 m. Number of tiles needed:
- (A) 40
 - (B) 20
 - (C) 9
 - (D) 18
82. If the body could not make enough red blood cells, the main problem would be:
- (A) no germ fighting
 - (B) blood won't clot
 - (C) not enough oxygen carried to the cells
 - (D) no urine made
83. The young plant inside a seed, which grows during germination, is the:
- (A) petal
 - (B) stigma
 - (C) ovary
 - (D) embryo

84. The length of a rectangle is 3 times its breadth. If the breadth is 4 cm, the perimeter is:
- (A) 24 cm
 - (B) 16 cm
 - (C) 48 cm
 - (D) 32 cm
85. Sweating helps the body in two ways: removing some wastes and:
- (A) digesting food
 - (B) making blood
 - (C) adding heat
 - (D) cooling the body as the sweat evaporates
86. Which floral part must pollen reach for fertilisation to begin?
- (A) the petal
 - (B) the sepal
 - (C) the root
 - (D) the stigma (top of the pistil)
87. The cost of carpeting a room 6 m by 5 m at ■50 per square metre is:
- (A) ■1500
 - (B) ■150
 - (C) ■300
 - (D) ■550
88. Which group correctly lists parts of the human circulatory system?
- (A) heart, blood, and blood vessels
 - (B) lungs, windpipe, nose
 - (C) stomach, liver, intestine
 - (D) kidney, ureter, bladder
89. A square and a rectangle have the same area of 36 cm². The square's side is 6 cm. A possible rectangle is:
- (A) 8 cm by 4 cm
 - (B) 9 cm by 4 cm
 - (C) 10 cm by 3 cm
 - (D) 6 cm by 5 cm
90. The tube that carries urine from each kidney to the urinary bladder is the:
- (A) windpipe
 - (B) artery
 - (C) ureter
 - (D) food pipe

91. Money plant grown from a stem piece kept in water is an example of:
- (A) vegetative propagation by stem cutting
 - (B) seed germination
 - (C) spore formation
 - (D) pollination
92. If the radius of a circle is doubled, its circumference becomes:
- (A) 4 times
 - (B) 2 times as large
 - (C) $\frac{1}{2}$ times
 - (D) the same
93. Which is TRUE about transport in plants vs animals?
- (A) animals use xylem
 - (B) plants have a heart
 - (C) neither transports anything
 - (D) plants use xylem and phloem; animals use a heart, blood and vessels
94. Which of these is NOT a method of asexual reproduction in plants?
- (A) growing from a cutting
 - (B) budding in yeast
 - (C) spore formation in fungi
 - (D) fertilisation of egg by pollen
95. The area of a rectangle is 60 cm^2 and its breadth is 5 cm. Its perimeter is:
- (A) 24 cm
 - (B) 65 cm
 - (C) 34 cm
 - (D) 17 cm
96. Which fluid transports oxygen, food and wastes around our body?
- (A) blood
 - (B) saliva
 - (C) bile
 - (D) air
97. The bright colour and sweet smell of many flowers mainly help to:
- (A) make food
 - (B) absorb water
 - (C) attract insects for pollination
 - (D) scare animals away
98. The area of a square of side 12 cm is:
- (A) 48 cm^2
 - (B) 144 cm
 - (C) 144 cm^2
 - (D) 24 cm^2

99. Plants lose water vapour to the air mainly through tiny leaf pores called:

- (A) stomata
- (B) xylem
- (C) roots
- (D) veins

100. After fertilisation, each ovule develops into a:

- (A) sepal
- (B) petal
- (C) seed
- (D) stigma